Phase 5 - Formal Report

1. Introduction

Our project involves investigating how movie discussions might relate to commercial success, using Reddit discussions and box office revenue as proxies for movie discussion and commercial success. By analyzing box office numbers and other movie specifications in relation to movie discussions—which itself does not have specific rating-based systems for recommendations—we aim to build a better understanding of what general movie audiences value.

Movies are important to cultivating culture and connecting audiences of all ages and backgrounds. With the rise of movie recommendation systems such as Letterboxd in the younger generation and the continued use of platforms such as IMBd, Rotten Tomatoes, and more, it is clear audiences value the relative popularity of the movies they are watching and want to watch. Reddit happens to be a huge source of movie recommendations, with the r/movie suggestions subreddit alone having over 800k+ weekly visitors. However, since Reddit does not have the same types of ranking metrics as other platforms such as IMDb, it is hard to determine how "good" or "successful" a recommended movie actually is. Since anyone can make a Reddit post, we want to understand whether the Reddit recommendations (measured through upvotes, or frequency) relate to a movie's commercial success. This helps us see whether the films most frequently suggested on Reddit are the same as the ones that are financially successful and are broadly enjoyed. In investigating genres, we will focus on horror, action, and comedy, as they may be best reflected through movie theatre revenue due to their theatrical or communal elements. In accordance with our research topic, our three questions are as follows:

1. How does Reddit recommendation frequency relate to the proportion of a movie's box office revenue earned internationally versus domestically among the movie genres horror, action, and comedy?
2. How does the average IMDb vote on a movie relate to the number of Reddit recommendations versus the box office success?
3. How does the average number of upvotes for Reddit recommendations and discussion posts vary by movie duration, and how does this relationship change based on movie genre (focusing on the genres of comedy, horror, and action)?

By examining the relationship between how often a movie is recommended on Reddit, its commercial performance, and other movie characteristics, we can better understand what audiences value in choosing movies to recommend. The findings from these questions will be valuable to film producers, directors, producers, and passionate film audiences. For movie production, an understanding of movie characteristics that facilitate more discussion among audiences may help shape decision-making surrounding a movie's features. Similarly, audiences who enjoy discussion of films may be interested in watching movies that share characteristics associated with films with high discussion frequency.

2. Data

The 3 datasets below were chosen to investigate what makes a movie successful and whether multiple metrics relate to one another.

Dataset 1 - IMDb: https://github.com/sahildit

The IMDb dataset contains 300,000+ movies ranging from release years 1894 to 2020, including titles, languages, and genres for each. The attributes we plan to use from this dataset are the title, genre, year, votes, avg_vote, and imdb_title_id. This dataset was chosen because it is useful for all three research questions, providing core movie information and allowing for the integration of the other datasets. The remaining datasets use differing movie identifiers, and this IMDb dataset contains both. Therefore, this dataset acts as a bridge for the 2 other datasets and provides key movie attributes.

This dataset was cleaned by first only keeping the columns needed for our analysis: imdb_title_id, title, year, genre, duration, avg_vote, votes, and language. We filtered the data to English-language movies released between 2010 and 2018, matching the time range of the Box Office Mojo dataset. We then restricted the data to movies whose genre list included at least one of horror, action, or comedy, since our research questions focus on these three genres. Next, we removed rows with missing values in key columns along with any duplicate rows. Lastly, to support joining with Box Office Mojo, we standardized the formatting of the movie titles

Dataset 2 - Reddit: https://huggingface.co/datasets/ZhankuiHe/reddit_movie_small_v1

This dataset contains Reddit movie recommendation discussions during the full year of 2022 from five subreddits (r/movies, r/moviesuggestions, r/bestofnetflix, r/netflixbestof, r/truefilm) and has been processed to identify movie mentions and link them to IMDb IDs. This dataset contains 420,000 rows, where each row represents one conversational turn within a recommendation dialogue. This dataset was chosen because we wanted to utilize an unconventional source for quantifying and analyzing movie success. The source is rich in data due to Reddit's large and active movie community. From this, we can calculate post counts and total and average upvotes per movie to compare against more common success metrics. This enables this dataset to support all three research questions.

For cleaning the Reddit dataset, we first kept only the attributes that are relevant to our research questions: turn_id, upvotes, is_seeker, and the processed text field containing IMDb IDs, which we stored as imdb_title_id. We dropped any rows that had null values in these fields. We then scanned the processed_id column for IMDb IDs and built a new table where each row represents a single movie mention. Finally, we removed duplicate rows that had the same combination of turn_id and imdb_title_id, so that repeated mentions of the same movie in a single post were not counted as separate recommendations.

Dataset 3 - Box Office Mojo:
https://github.com/ntdoris/movie-revenue-analysis/blob/main/zippedData/bom.movie_gross.csv.gz
This dataset from Box Office Mojo contains data from approximately 4000 movies, with release dates ranging from 2010 to 2018. Key attributes in this dataset consist of domestic_gross, which refers to the combined Canada and USA box office gross, and foreign_gross, which refers to the combined box office gross from the rest of the world. This dataset was chosen to supplement the missingness in the IMDb dataset's box office revenue. These metrics were utilized to answer research questions 1 and 2.

In cleaning the Box Office Mojo dataset, we kept only the movie title, year, domestic_gross, and foreign_gross. We renamed the movie column to title and standardized it in the same way as in IMDb so that the two datasets could be joined on title and year. We converted year, domestic_gross, and foreign_gross to numeric types, then dropped rows missing a title or year, and removed rows where both domestic and foreign gross were missing or both were negative values. As a final step, we removed duplicate (title, year) pairs so that there is at most one Box Office Mojo record per movie in the cleaned file.

During EDA of the raw IMDb dataset, we processed the genre attribute by splitting the comma-separated genre strings into lists, and then exploded them so that each genre appeared as a separate row for each movie. This allowed us to accurately count the frequency of each genre without treating multi-genre combinations as a single category. We selected horror, action, and comedy as our genres of interest because they were among the most frequently occurring genres in the dataset, ensuring adequate sample sizes for meaningful statistical analysis. Using an inspection script, we also examined the data types, unique values per column, and percentage of missing values across variables in all 3 datasets to assess feasibility for analysis. Variables with high amounts of missing values or low uniqueness, such as budget and meta score, were excluded. This allowed us to narrow down our research questions to focus on features with sufficient and reliable data. For each dataset, we also performed manual spot checks to verify that the information matched correctly with the official IMDb, Reddit, and Box Office Mojo websites. Spot checks were also done comparatively between the datasets, verifying that the titles, years, and IDs matched correctly across each other, ensuring our joins and cleaning steps behaved as expected.

The most significant limitation is the Box Office Mojo dataset's narrow scope, which limits the analysis to movies released in 2010-2018. We also focus only on English movies in the genres of horror, action, or comedy, introducing bias into the analysis; therefore, the findings may not generalize well. The Reddit dataset also contains bias because all the data was collected from 2022 Reddit postings, and would thus only reflect films that were popular or trending in 2022.

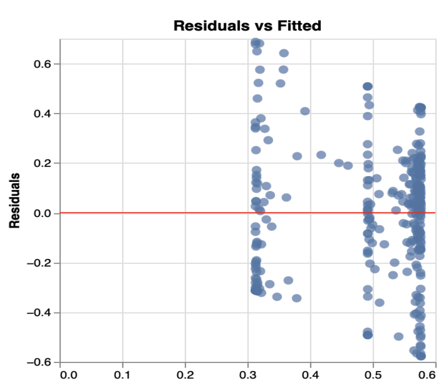
Additionally, the Reddit dataset only covers posts from specific movie-related subreddits and only includes users who choose to post there, so it may over-represent highly engaged or niche communities. Lastly, due to noisy raw text and an imperfect automated processor, the Reddit data may contain errors.

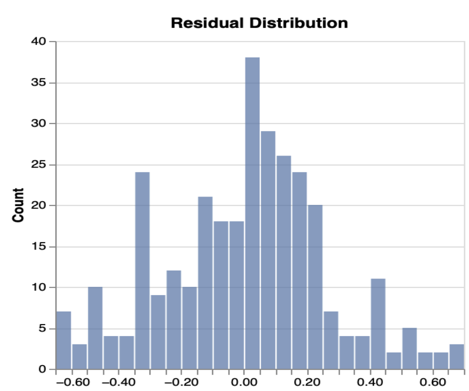
3. Methodology

To analyze the relationship between Reddit recommendation frequency and the proportion of box office earned internationally and domestically, we calculated the following variables for each movie: number of Reddit recommendations, foreign revenue share, and domestic revenue share. Because movies were associated with multiple genres in earlier phases, we modified our query to extract a movie if the primary genre (first genre in the list of genres) matches horror, action, or comedy. This modification enabled regression analysis. Using the query output, we created scatterplots to visualize the relationship between Reddit recommendation counts and both domestic and international revenue shares across genres. We then computed the Pearson correlation coefficient both overall and within each genre to assess the strength and direction of the linear relationship. The overall correlation provides a baseline measure across all movies, while genre-specific correlations capture variation across genres. To further analyze this relationship while controlling for genre, we conducted a multiple linear regression with interaction terms: Foreign Share =

$$\beta_0 + \beta_1(\mathit{Recommendations})_i + \beta_2(\mathit{Genre})_i + \beta_2(\mathit{Recommendations} \times \mathit{Genre}_i) + \epsilon_i .$$

This model allows us to examine both the overall effect of recommendation frequency and whether this effect differs across genres. Model assumptions were assessed using diagnostic plots. The histogram of residuals was approximately symmetric and centered around zero, supporting normality. The residuals vs. fitted values showed no clear pattern, suggesting linearity, and mild heteroskedasticity (indicated by the fanning) is observed, but it is not severe. From the regression output, Durbin-Watson statistic was close to 2, indicating the errors are independent.





For question 2, we used database queries to extract movie-level data, including the average IMDb rating, total box office gross, and Reddit discussion count. We created scatterplots with fitted regression lines to visualize relationships between IMDb and each explanatory variable.

We then conducted simple and multiple linear regressions, regressing IMDb rating on (1) total gross, (2) Reddit discussion count, and (3) both variables jointly. Simple regressions isolate individual relationships, while the multiple linear regression assesses their combined effects while controlling for the influence of the other variable. Model performance was evaluated using R^2 , which is interpreted as the proportion of variance in average IMDb votes explained by its predictors.

For question 3, we unwound the reddit_mentions and genre fields to analyze each recommendation individually and filtered the dataset to include only horror, action, and comedy. We then created duration bins (eg., 60-79, 80-89 minutes) to group movies into meaningful intervals. Finally, we grouped the data by both duration bin and genre, and the average number of upvotes was computed for each combination. The results were visualized using a line chart to compare how upvotes change across duration for each genre. Although we initially planned to use ANOVA to test for differences across groups, we determined that the aggregated results and visualizations were sufficient enough to illustrate the patterns without formal statistical testing.

4. Database Implementation

For the Oracle implementation, we designed a relational schema with the following tables: IMDB_MOVIES, BOM_GROSS, and REDDIT_MENTIONS. IMDB_MOVIES stores the core movie attributes and uses imdb_title_id as the primary key. REDDIT_MENTIONS references IMDB_MOVIES through a foreign key on imdb_title_id, capturing the one-to-many relationship between a movie and its associated Reddit discussion posts. Since Reddit mentions are identified within the context of a movie, the combination of imdb_title_id and turn_is uniquely identifies each record, meaning REDDIT_MENTIONS can be viewed as a weak entity independent of IMDB_MOVIES. BOM_GROSS stores domestic and foreign revenue information separately and is linked to movies using title and year. As the Box Office Mojo dataset does not provide a unique identifier, the combination of title and year serves as the identifying attributes for BOM_GROSS records in our schema. A key decision was normalization into separate tables to reduce redundancy and maintain data integrity. For example, Reddit mentions were stored separately rather than duplicated within movie records, ensuring each mention is stored exactly once and can be independently queried. Foreign key constraints further enforce referential integrity by ensuring all Reddit mentions correspond to a valid movie.

For the MongoDB implementation, we used a single collection called movies, where each document represents one movie. The top-level fields include imdb_title_id, title, year, genre, duration, avg_vote, votes, and language. Box office data is embedded as a subdocument with domestic_gross and foreign_gross. Reddit discussion is stored as an embedded array containing turn_id, upvotes, and is_seeker. This data was embedded because box office data and Reddit mentions are consistently accessed together with movie data in our analysis, making each document self-contained, eliminating joins, and simplifying queries.

As the relational and document-oriented databases use different modelling approaches, several design decisions differed. In Oracle, data is normalized into separate tables and relationships are reconstructed through joins. In contrast, MongoDB emphasizes denormalization, embedding data that would be stored in separate tables in Oracle. Relationships are explicitly defined using foreign keys in Oracle, whereas in MongoDB, they are represented through document structure.

For this dataset, the MongoDB approach felt more natural because our analysis focused on each movie as a single unit of analysis containing its metadata, box office results, and Reddit activity. This single document representation is aligned closely with how we queried and visualized the data. In contrast, the Oracle implementation required multiple joins to reconstruct the same view, making queries more complex and less intuitive for our specific research questions. While MongoDB simplified query construction, the tradeoff was some data duplication within documents, whereas Oracle aimed to reduce redundancy through normalization. However, due to limitations in the Box Office Mojo dataset, such as non-unique title and year combinations, some duplications still could arise when joining tables.

5. Database Comparison and Reflection

Overall, we would choose MongoDB for this project as it allowed for the structuring of the data so that each movie was treated as a single document with associated metadata, box office performance, and Reddit activity. This method of data representation reduced the need for complex joins in our queries. In contrast, the SQL queries are more complex due to the required joins and grouping logic, and the discrepancy we encountered in question 1 highlights this difference in how the two databases handle data structure and relationships.

To make Oracle more preferable over MongoDB, the required changes would depend on the question. For question 2, by grouping using imdb_title_id and calculating the count in a separate query and replacing the Reddit data in the main query with the query result would prevent a bloated GROUP BY function, allowing the focus to stay on the calculations. The other two questions would benefit from each movie entry only having one associated genre, as this would make it easier to sort and group by genre. For question 3, given this change and the simplicity of binning movie durations in SQL, Oracle would be the preferable choice.

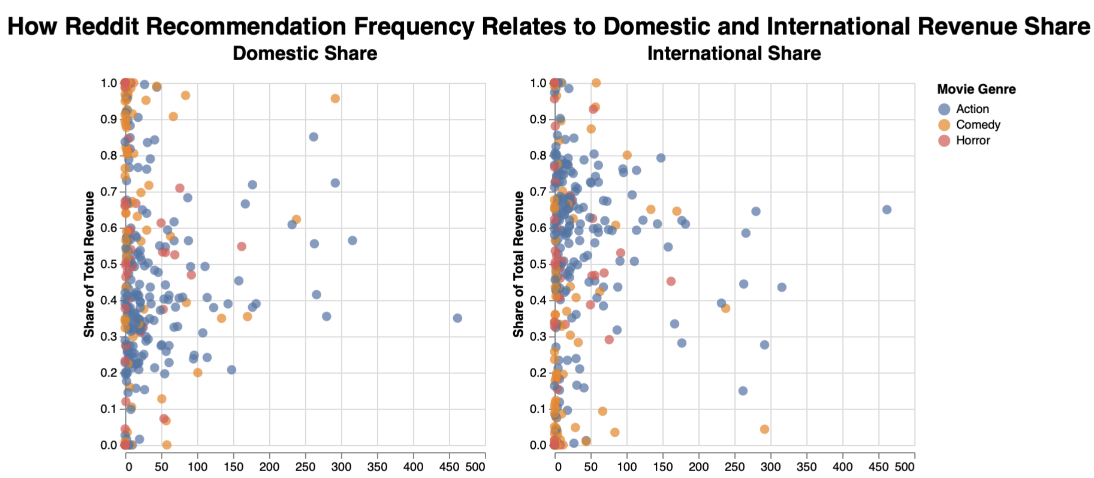
It was easier to write the queries for question 1 in MongoDB because the Reddit mentions and box office data were embedded within each movie document, so we could compute variables like recommendation counts and foreign revenue share without joins. In contrast, the SQL queries were more complex due to the joins and grouping logic, and the discrepancy we encountered in the question highlights this difference in databases. Initially, the SQL query returned 34 extra rows due to duplicate (title, year) combinations in the box office data. This made the SQL query more error-prone and required us to use DISTINCT. In MongoDB, this issue did not arise because each document is uniquely identified with imdb_title_id, ensuring one record per movie. After correcting the SQL query using DISTINCT, both systems produced consistent results.

For question 2, the MongoDB query was significantly easier to write compared to the SQL query. Using MongoDB, there was no need to use any grouping operators, since the MongoDB loading code already aggregated everything by movie. This allowed for reddit_discussion_count to be calculated directly by using $size on the array made in the data loading code containing all Reddit mentions of the movie. In contrast, the SQL query was more difficult to write. Since the variables such as avg_vote, domestic_gross, and foreign_gross had a one-to-one relationship with a single movie, they had to be included in the GROUP BY function to be utilized properly.

MongoDB made the query for question 3 easier to write, as the $unwind operator allowed the genre array to be expanded into separate documents, allowing movies with multiple genre groups to contribute to the calculation of average upvotes for more than one genre category. In SQL, the approach involved replacing each list of genres with a single value containing the first relevant genre, as grouping by genre would not be feasible if values remained as a list. Different genre combinations (eg., comedy and romance vs. comedy-drama) would be treated as distinct groups despite sharing a common genre of interest. However, we later realized that this approach caused movies with multiple relevant genres to be counted in only one category. Troubleshooting the query was easier in MongoDB Compass, as using the aggregation pipelines allowed us to see immediate outputs at each stage. However, binning the movie durations was easier in SQL, as all the bins could be created within a single CASE WHEN block, while MongoDB required multiple $andFields and $cond stages to construct the same bins, making the pipeline longer.

6. Results

<u>Question 1: How does Reddit recommendation frequency relate to the proportion of a movie's box office revenue earned internationally versus domestically among the movie genres horror, action, and comedy?</u>



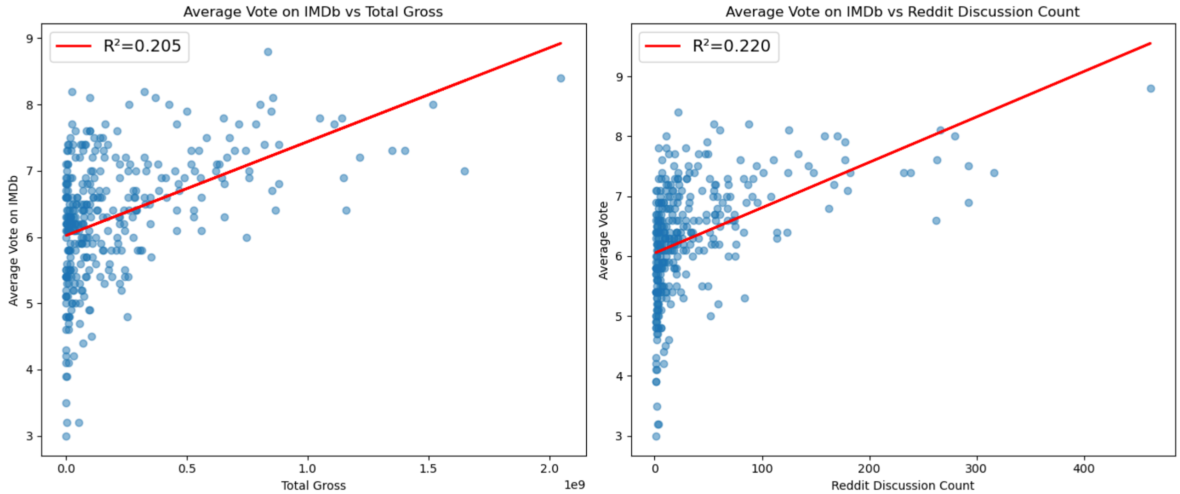
```
                            OLS Regression Results
==============================================================================
Dep. Variable:          foreign_share   R-squared:                       0.134
Model:                            OLS   Adj. R-squared:                  0.120
Method:                 Least Squares   F-statistic:                     9.638
Date:                Fri, 03 Apr 2026   Prob (F-statistic):           1.44e-08
Time:                        08:52:18   Log-Likelihood:                -23.741
No. Observations:                 317   AIC:                             59.48
Df Residuals:                     311   BIC:                             82.04
Df Model:                           5
Covariance Type:            nonrobust
================================================================================================================
                                                coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------------------------
Intercept                                     0.5771      0.022     26.619      0.000       0.534       0.620
C(primary_genre)[T.Comedy]                   -0.2642      0.039     -6.753      0.000      -0.341      -0.187
C(primary_genre)[T.Horror]                   -0.0852      0.054     -1.590      0.113      -0.191       0.020
num_recommendations                          -0.0003      0.000     -0.889      0.375      -0.001       0.000
num_recommendations:C(primary_genre)[T.Comedy]  0.0010    0.001      1.576      0.116      -0.000       0.002
num_recommendations:C(primary_genre)[T.Horror]  0.0003    0.001      0.237      0.813      -0.002       0.003
==============================================================================
Omnibus:                        0.008   Durbin-Watson:                   1.878
Prob(Omnibus):                  0.996   Jarque-Bera (JB):                0.055
Skew:                          -0.011   Prob(JB):                        0.973
Kurtosis:                       2.939   Cond. No.                         262.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

	genre	correlation
0	Overall	0.029798
1	Action	-0.071855
2	Comedy	0.123090
3	Horror	0.005825

The scatterplots illustrate the relationship between Reddit recommendation frequency and both domestic and international revenue shares across horror, action, and comedy films. Most movies have fewer than 50 recommendations. International revenue share is most commonly concentrated between 0.4 and 0.8, though variation exists outside this range, and does not appear to change, regardless of Reddit recommendation frequency. Since domestic revenue share is the complement of international share (1 − international share), this same lack of relationship with recommendation frequency is observed for domestic revenue as well. However, action films appear more frequently at higher recommendation counts and tend to have higher international revenue shares, while comedy and horror films are more concentrated at lower recommendation counts.

The overall Pearson correlation is 0.03, indicating there is almost no linear relationship. When broken down by genre, comedy shows a small positive correlation (0.12), action shows a slight negative correlation (-0.07), and horror is near zero (0.006). These values suggest that the relationship between recommendation frequency and revenue share is negligible across all genres. The multiple linear regression results are consistent with this finding as the coefficients for recommendation frequency and its interaction with genre are small and statistically insignificant. The R^2 value of 0.13 indicates that the model explains very little of the variation in revenue share. Although we hypothesized that higher Reddit recommendation frequency would be associated with higher commercial success, especially for horror films, there is little to no relationship between Reddit recommendation frequency and the proportion of box office revenue earned internationally versus domestically. More specifically, the horror genre does not exhibit a stronger association between Reddit recommendation frequency and revenue share in either market, contradicting our expectation that its theatrical appeal would produce a stronger relationship. While the visualizations and statistical outputs all suggest Reddit discussion is not a strong indicator of revenue distribution, other factors such as marketing or global audience preferences may play a larger role.

<u>Question 2: How does the average IMDb vote on a movie relate to the number of Reddit recommendations versus the box office success?</u>



To examine how IMDb average rating relates to Reddit discussion and box office success, three regression models were estimated: (1) Average Vote vs. Total Gross, (2) Average Vote vs Reddit Discussion Count, and (3) Average Vote vs. Total Gross and Reddit Discussion Count. R^2 was used as the evaluation metric due to its interpretability in measuring the proportion of variance explained.

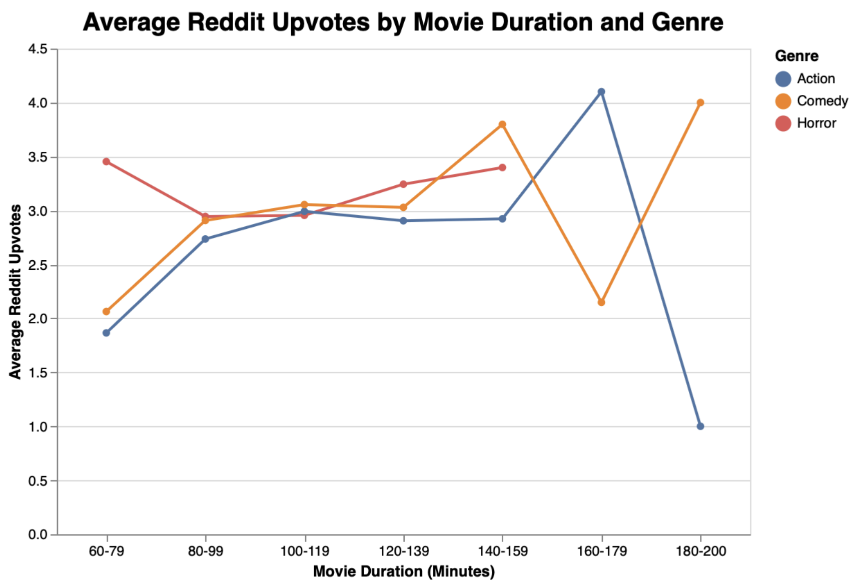
The small multiples display the first two single linear regressions, along with their fitted trend lines. Each point represents a movie, and both plots show strong clustering at low values of Total Gross and Reddit Discussion Count. These clusters are centred around an Average Vote of 6. We did not model the multiple linear regression because it has two predictors and models a 3D relationship, which is difficult to cleanly show in a single 2D scatterplot. However, the multiple linear regression results in an R^2 score of 0.404.

Both relationships are positive but weak. The model using Total Gross achieves an R^2 of 0.205, while the model using Reddit Discussion Count has an $R^2$of 0.220, indicating that, individually, neither variable explains much variation in IMDb ratings. Our initial hypothesis was that these 2 variables would have a strong, positive relationship with Average Vote, assuming that the two variables could give valuable information to a movie's IMDb average vote. These graphs support our hypothesis that the two relationships are positive, but go against our hypothesis in terms of predicted relationship strength.

However, when both variables are included in a multiple linear regression, the R^2 increases to 0.404. This suggests that while total gross and Reddit discussion count are weak predictors on their own, they jointly explain a moderate portion of the variance in IMDb ratings. This suggests that there is a relationship between the average vote and the combination of Reddit Discussion Count and Box Office Success. This could be explained by the two variables being different measures of movie success, so perhaps combining them gives a more complete picture of what influences IMDb ratings.

To answer this research question, despite neither Reddit Discussion Count nor Box Office Success having strong positive relationships with the Average Vote of a movie on IMDb, combining the two features suggests that the two variables are jointly responsible for a moderate amount of a movie's average IMDB vote.

<u>Question 3: How does the average number of upvotes for Reddit recommendations and discussion posts vary by movie duration, and how does this relationship change based on movie genre (focusing on the genres of comedy, horror, and action)?</u>



genre	duration_bin	avg_upvotes_SQL	avg_upvotes_mongodb
Action	180-200	1.000000	1.000000
Action	60-79	1.875000	1.865385
Comedy	60-79	2.120219	2.063291
Comedy	160-179	2.148148	2.148148
Action	80-99	2.780497	2.735610

First six rows of query output, sorted by increasing average upvotes.

To answer this question, SQL and MongoDB queries were used to group the data by genre and newly created binned movie durations. For each genre-duration combination, the average number of Reddit upvotes was computed by aggregating upvotes across all associated Reddit posts. The query's output includes the average number of Reddit upvotes for each grouping.

The output visualization is a line chart that plots how the average number of Reddit upvotes varies across different movie duration bins for action, comedy, and horror films. The lines highlight trends across duration intervals, while colour is encoded to enable comparison between genres. For comedy and action films, there is generally a slight increase in the average number of upvotes as movie duration increases, up until the longer duration movies, where upvote averages diverge. Horror movies exhibit a different pattern, beginning with relatively high average upvotes for shorter durations, decreasing in the next bin, and gradually increasing again, before the line ends due to a lack of data. This limits interpretation, as it is unclear whether it is due to an absence of Reddit discussion posts for movies that fall into these groups or an absence of movies in these groups entirely. Overall, while longer movie durations appear to be associated with higher Reddit upvotes for action and comedy films in the mid-range durations, this relationship is not consistent across all durations or genres. The line chart also suggests that the effect of movie duration on audience engagement varies by genre.

The ANOVA analysis originally planned for this question would have allowed us to formally test whether differences in average upvotes across genre and duration bins are statistically significant. However, as our analysis focuses on identifying general patterns and trends, and the visualization and query output already reveal differences across groups, we determined that conducting the additional ANOVA analysis was unnecessary. However, we cannot be certain of the statistical significance of our observations.

The findings from this query and visualization partially contradict the original hypothesis, which predicted a consistent positive relationship between movie duration and Reddit upvotes. This hypothesis was due to the idea that longer movies could facilitate the development of more complex plot points, generating more potential areas for discussion. While there is a slight increase in the average number of upvotes in the mid-upper range bins across genres (consistent with the hypothesis), the hypothesis is not consistent for the higher ranges of movie durations, nor for the horror genre at the lowest duration bin.

7. Limitations and Future Work

A key limitation of our study was how the genre data was stored as a list of genres, as each movie could belong to more than one category. To run the regression for question 1, we simplified this by selecting a movie if the first genre in the list of genres matches horror, action or comedy. However, this simplification introduces a classification bias because the assigned genre may fail to represent the full range of genres associated with the movie. Misclassifying the movies' genres may lead to biased coefficient estimates. Our queries for the second question removed movies with zero Reddit discussion, which excludes less-discussed or underground movies, which are likely to have lower revenue and engagement. As a result, the sample is skewed towards more popular films. It is worth noting that although selection biases typically inflate the observed relationships, the model yields a low R^2, reinforcing the conclusion that these variables have limited explanatory power. While writing the query for question 3 in MongoDB, we unwound the genre field so each movie could be analyzed within every genre to ensure we did not lose information. However, this means the same movie can contribute to multiple categories, meaning the observations are not independent. Additionally, handling of multi-genre movies differs between SQL and MongoDB: in SQL, each movie is assigned to a single genre, while in MongoDB, the same movie can appear in multiple genre groups after using unwind, which can slightly change aggregated averages. Another limitation comes from how the datasets are joined. Box Office Mojo identifies movies using title and year, while Reddit uses IMDb IDs. We used the IMDb dataset as a bridge between them, but even after standardizing titles, there can still be ambiguous or conflicting matches when movies share similar titles or when titles are formatted differently. In Phase 4, we found that relying on a (title, year) combination in SQL led to duplicate movies, while the MongoDB collection, which uses imdb_title_id as the main key, stored each movie only once. This shows that our analysis can be sensitive to how we define and match movie identities.

With more time and resources, we would add more features such as budget, franchise status, and more detailed marketing or release information to control for factors that likely affect both revenue and online discussion. We would also incorporate more subreddits or other social media platforms, and use sentiment analysis on Reddit posts to separate positive and negative discussions. To address the multiple genres a movie belongs to, we could explore multi-label modelling techniques to better account for the multiple genres that movies can belong to. Including all genres a movie belongs to would reduce misclassification and better capture the complexity of genre effects on revenue and engagement.

From the first research question, we concluded that there is little to no relationship between Reddit recommendation frequency and how the proportion of revenue is split between domestic and international markets across all genres. This raises the question of whether factors such as global audience preferences and marketing play a larger role in determining how box office revenue is split between domestic and international markets. The results of research question 3 also raises questions around why some genres may be associated with different magnitudes of upvotes in Reddit discussions. Our findings also raise questions about how the timing of online discussions relative to a movie's release date affects box office performance, and whether certain studios or franchises consistently generate higher engagement for the same level of commercial success.

wait sorry do you understand what i mean

8. References

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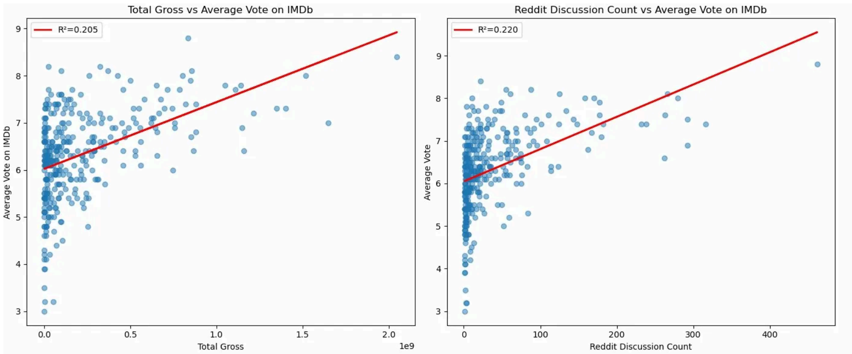
9. AI Tool Use Declaration

Tool: Claude

Prompt: How to make the font displaying the r2 scores bigger?



Answer:

Just change the `fontsize` in the legend:

```python
ax1.legend(fontsize=14)
ax2.legend(fontsize=14)
```

Adjust the number up or down to whatever size you want.